

12. (g) Implementation Plan: B.Tech. in Biomedical Engineering

Semester – I						Semester – II					
Code	Course Title	Hours/Week			C	Code	Course Title	Hours/Week			C
		L	T	P				L	T	P	
21LEH101T	Communicative English	2	1	0	3	21LEH102T	Chinese				
21MAB101T	Calculus and Linear Algebra	3	1	0	4	21LEH103T	French				
21PYB101J	Physics: Electromagnetic Theory, Quantum Mechanics, Waves and Optics	3	1	2	5	21LEH104T	German	2	1	0	3
21MES102L	Engineering Graphics and Design	0	0	4	2	21LEH105T	Japanese				
21EES101T	Electrical and Electronics Engineering	3	1	0	4	21LEH106T	Korean				
21BMC101J	Biomedical Sensors	2	0	2	3	21LEH107T	Spanish				
21CYM101T	Environmental Science	1	0	0	0	21GNH101J	Philosophy of Engineering	1	0	2	2
21PDM101L	Professional Skills and Practices	0	0	2	0	21MAB102T	Advanced Calculus and Complex Analysis	3	1	0	4
21LEM101T	Constitution of India	1	0	0	0	21CYB101J	Chemistry	3	1	2	5
Total Credits					21	21CSS101J	Programming for Problem Solving	3	0	2	4
						21BTB104T	Biology-Human Physiology and Anatomy	2	0	0	2
						21MES101L	Basic Civil and Mechanical Workshop	0	0	4	2
						21PDM102L	General Aptitude	0	0	2	0
						21GNM101L	Physical and Mental Health using Yoga				
						21GNM102L	NSS	0	0	2	0
						21GNM103L	NCC				
						21GNM104L	NSO				
						Total Credits					22
Semester – III						Semester – IV					
Code	Course Title	Hours/Week			C	Code	Course Title	Hours/Week			C
		L	T	P				L	T	P	
21MAB201T	Transforms and Boundary Value Problems	3	1	0	4	21MAB202T	Numerical Methods	3	1	0	4
21BMC202T	Biomedical Signals and Systems	3	0	0	3	21CSE206T	Artificial Intelligence	2	1	0	3
21BMC203J	Electric and Electronic Circuits	3	0	2	4	21BMC205J	Integrated Circuit Design for Bioinstrumentation	2	0	2	3
21BMC204J	Digital Logic for Medical Systems	2	0	2	3	21BMC206J	Biomedical Instrumentation	3	0	2	4
21PDH201T	Social Engineering	2	0	0	2	21BMC207J	Biomaterials and Tissue Interaction	2	0	2	3
21PYS202T	Medical Physics	3	0	0	3	E	Professional Elective – I				3
21LEM201T	Professional Ethics	1	0	0	0	21DCS201P	Design Thinking and Methodology	1	0	4	3
21PDM201L	Verbal Reasoning	0	0	2	0	21PDM202L	Critical and Creative Thinking Skills	0	0	2	0
21LEM202T	Universal Human Values– Understanding Harmony and Ethical Human Conduct	2	1	0	3	Total Credits					23
Total Credits					22						
Semester – V						Semester – VI					
Code	Course Title	Hours/Week			C	Code	Course Title	Hours/Week			C
		L	T	P				L	T	P	
21MAB301T	Probability and Statistics	3	1	0	4	21CSS303T	Data Science	1	1	0	2
21BMC301J	Biomedical Signal Processing	3	0	2	4	21BMC304J	Medical Image Processing	2	0	2	3
21BMC302J	Microcontrollers and its Application in Medicine	3	0	2	4	21BMC305T	Biocontrol Systems	3	0	0	3
21BMC303T	Principles of Medical Imaging	3	0	0	3	E	Professional Elective – III				3
E	Professional Elective – II				3	E	Professional Elective – IV				3
O	Open Elective – I				3	21BMP303T	MOOC	3	0	0	3
21PDM301L	Analytical and Logical Thinking Skills	0	0	2	0	21BMP302L	Project	0	0	6	3
21LEM301T	Indian Art Form	1	0	0	0	O	Open Elective – II				3
21GNP301L	Community Connect	0	0	2	1	21PDM302L	Employability Skills and Practices	0	0	2	0
Total Credits					22	21LEM302T	Indian Traditional Knowledge	1	0	0	0
						Total Credits					20
Semester – VII						Semester – VIII					
Code	Course Title	Hours/Week			C	Code	Course Title	Hours/Week			C
		L	T	P				L	T	P	
21GNH401T	Behavioral Psychology	2	0	2	3	21BMP401L	Major Project	0	0	30	15
E	Professional Elective – V				3	21BMP402L	Major Project	0	0	20	
E	Professional Elective – VI				3	21BMP403L	Internship#	0	0	10	
21BMC401J	Biomechanics	2	0	2	3	Total Credits					15
21BMC402J	Biomedical Equipments for Clinical Applications	2	0	2	3						
O	Open Elective – III				3						
Total Credits					18						

#Students have to register either 21BMP401L or 21BMP402L and 21BMP403L both in eighth semester